Preliminary Report For Project 7
Sentiment Analysis

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Sentiment analysis is a key task in Natural Language Processing (NLP), aiming to automatically determine the emotional polarity of a given text. In this work, the goal is to address binary sentiment classification of Amazon product reviews. We distinguish positive (4–5 stars) from negative (1–2 stars) review. We use the Amazon Reviews dataset (Kaggle, Adame Bittlingmayer), which includes approximately 4 million labeled reviews. We propose to compare three approaches of increasing complexity: Logistic Regression with TF-IDF features, a Bi-LSTM recurrent neural network, and a fine-tuned DistilBERT transformer model. Models are evaluated using accuracy and macro F1-score on a held-out test set.

1 Introduction

2 Dataset

2.1 Description

2.2 Data Balance & Splits

2.3 Preprocessing

2.4 Evaluation Metrics

3 Solution Description

4 Conclusion

References

[1] NeurIPS (2024) Call for Papers
`https://neurips.cc/Conferences/2024/CallForPapers`

[2] Adam Bittlingmayer (2020) Amazon Reviews for Sentiment Analysis
`https://www.kaggle.com/datasets/bittlingmayer/amazonreviews`

A Appendix / supplemental material

Optionally include supplemental material (complete proofs, additional experiments and plots) in appendix. All such materials **SHOULD be included in the main submission.**

Submitted to 38th Conference on Neural Information Processing Systems (NeurIPS 2024). Do not distribute.